>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

MODELING OF PENDULUM MDOF STRUCTURE USING OPENSEES

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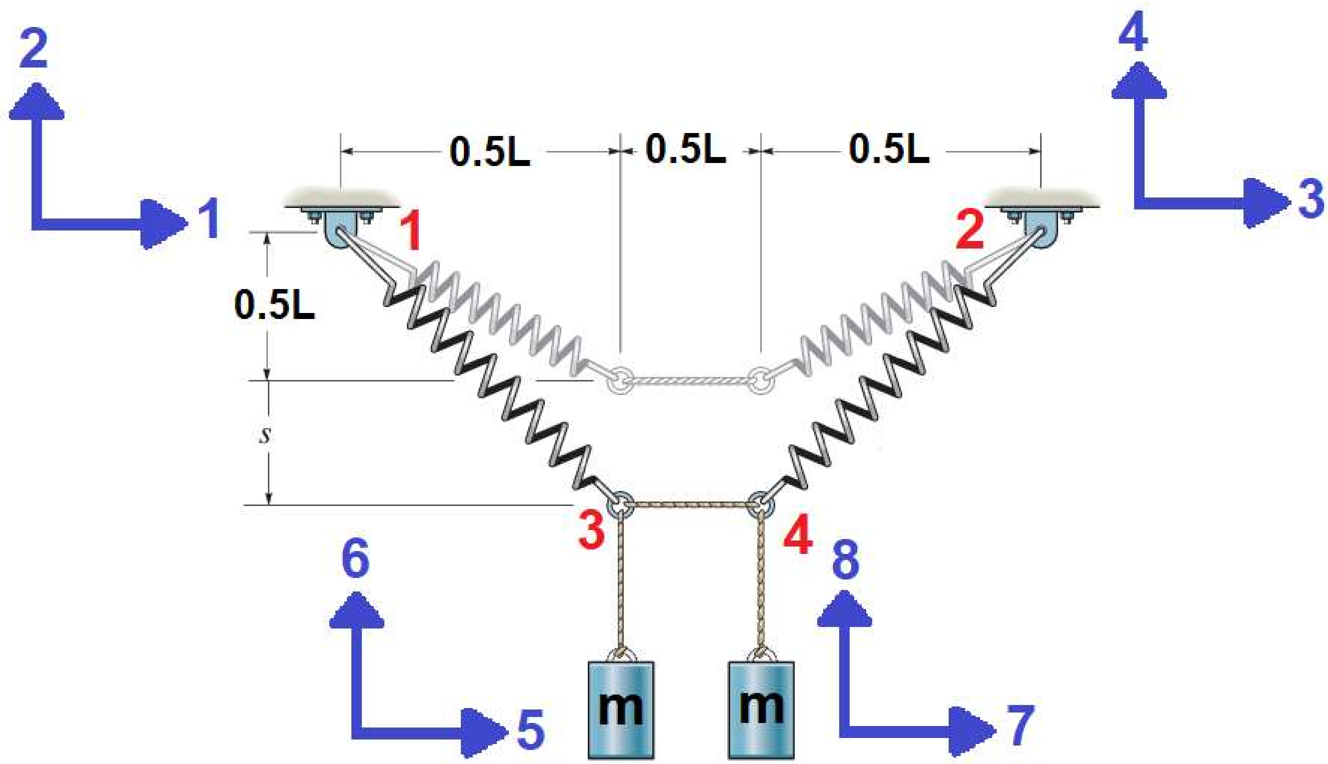
Spyder (Python 3.12)

C:\Users\Dell\Desktop\OPENSEES_FILES\+PENDULUM\+EXAMPLE_09\MDOF_PENDULUM__NINE.py

MDOF_PENDULUM__NINE.py

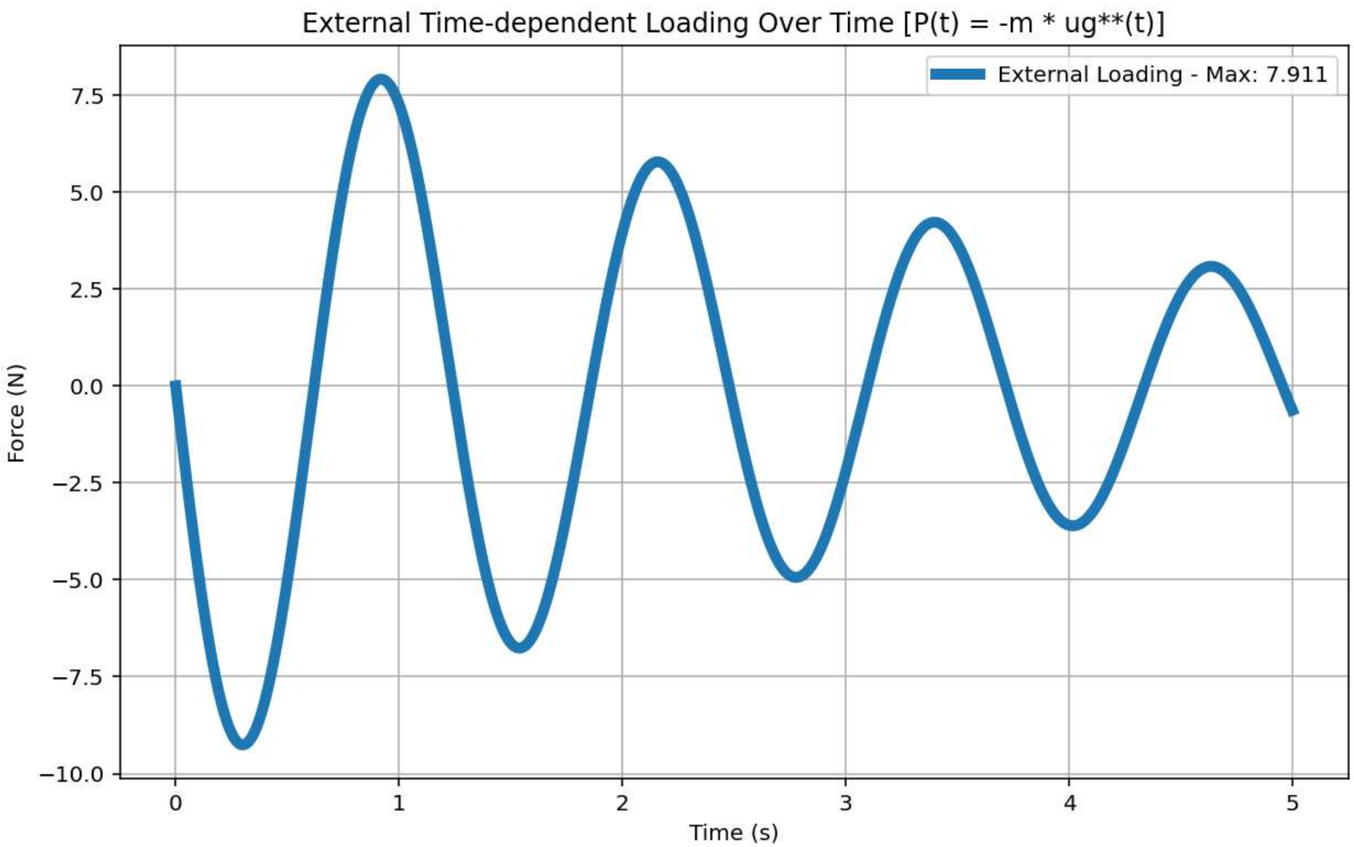
```python
#printModel()
ops.printModel("-JSON", "-file", "MDOF_PENDULUM_NINE.json")
#%%------------------------------------------------------------------------------
# EXCEL OUTPUT
import pandas as pd

# Create DataFrame function
def create_df(dispXE, dispXI, velXE, velXI, accelXE, accelXI, reactionXE, reactionXI, FDXe,
    df = pd.DataFrame({
        "DISPLACEMENT [m] - ELASTIC": dispXE,
        "VELOCITY [m/s] - ELASTIC": velXE,
        "ACCELERATION [m/s^2] - ELASTIC": accelXE,
        "BASE-REACTION [N] - ELASTIC": reactionXE,
        "DAMPING-FORCE [N] - ELASTIC": FDXe,
        "INERTIA-FORCE [N] - ELASTIC": FIXe,
        "SPRING-FORCE [N] - ELASTIC": FSXe,
        "DISPLACEMENT [m] - INELASTIC": dispXI,
        "VELOCITY [m/s] - INELASTIC": velXI,
        "ACCELERATION [m/s^2] - INELASTIC": accelXI,
        "BASE-REACTION [N] - INELASTIC": reactionXI,
        "DAMPING-FORCE [N] - INELASTIC": FDXi,
        "INERTIA-FORCE [N] - INELASTIC": FIXi,
        "SPRING-FORCE [N] - INELASTIC": FSXi,
    })
    return df


# Save to Excel
with pd.ExcelWriter("MDOF_PENDULUM_NINE_OUTPUT.xlsx", engine='openpyxl') as writer:

    df1 = create_df(dispXE, dispXI, velXE, velXI, accelXE, accelXI, reactionXE, reactionXI,
    df1.to_excel(writer, sheet_name="OUTPUT", index=False)
#%%------------------------------------------------------------------------------
```

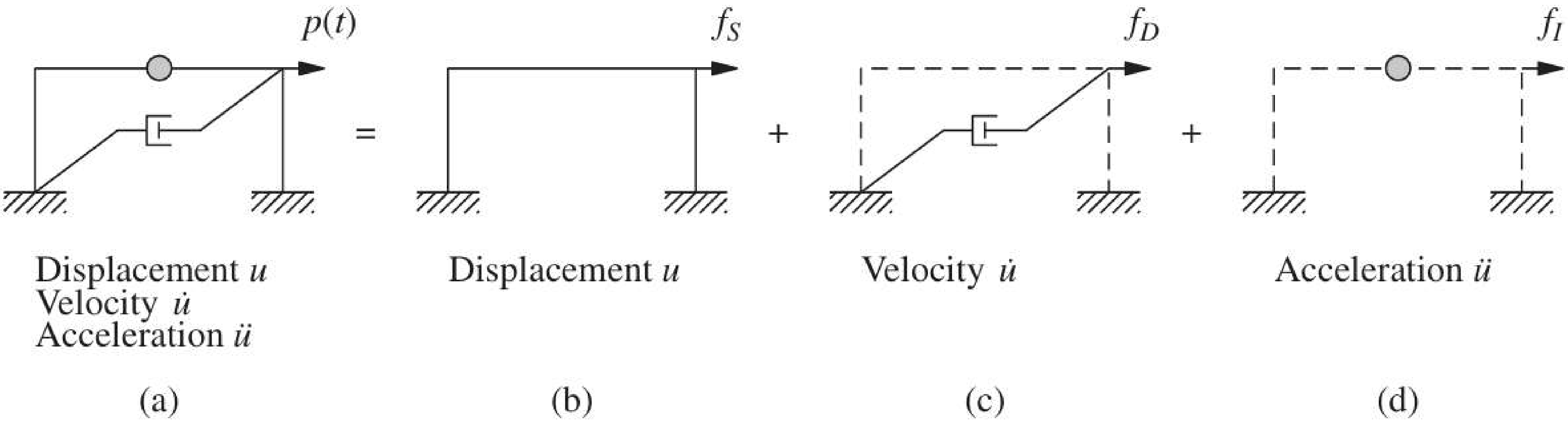
Conda: anaconda3 (Python 3.12.7) LSP: Python Line 594, Col 53 UTF-8 CRLF RW Mem 36%



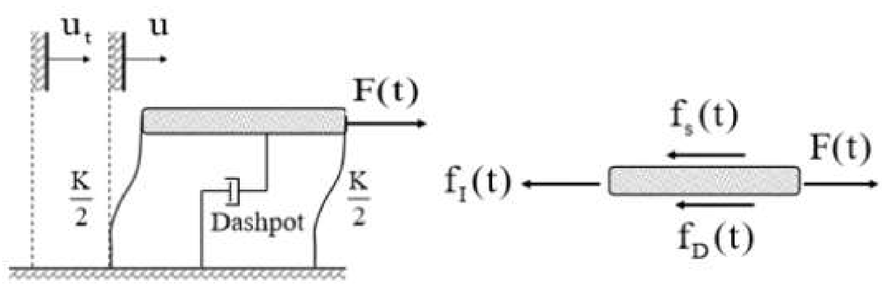
$$K_{\Delta}(P - \Delta)$$

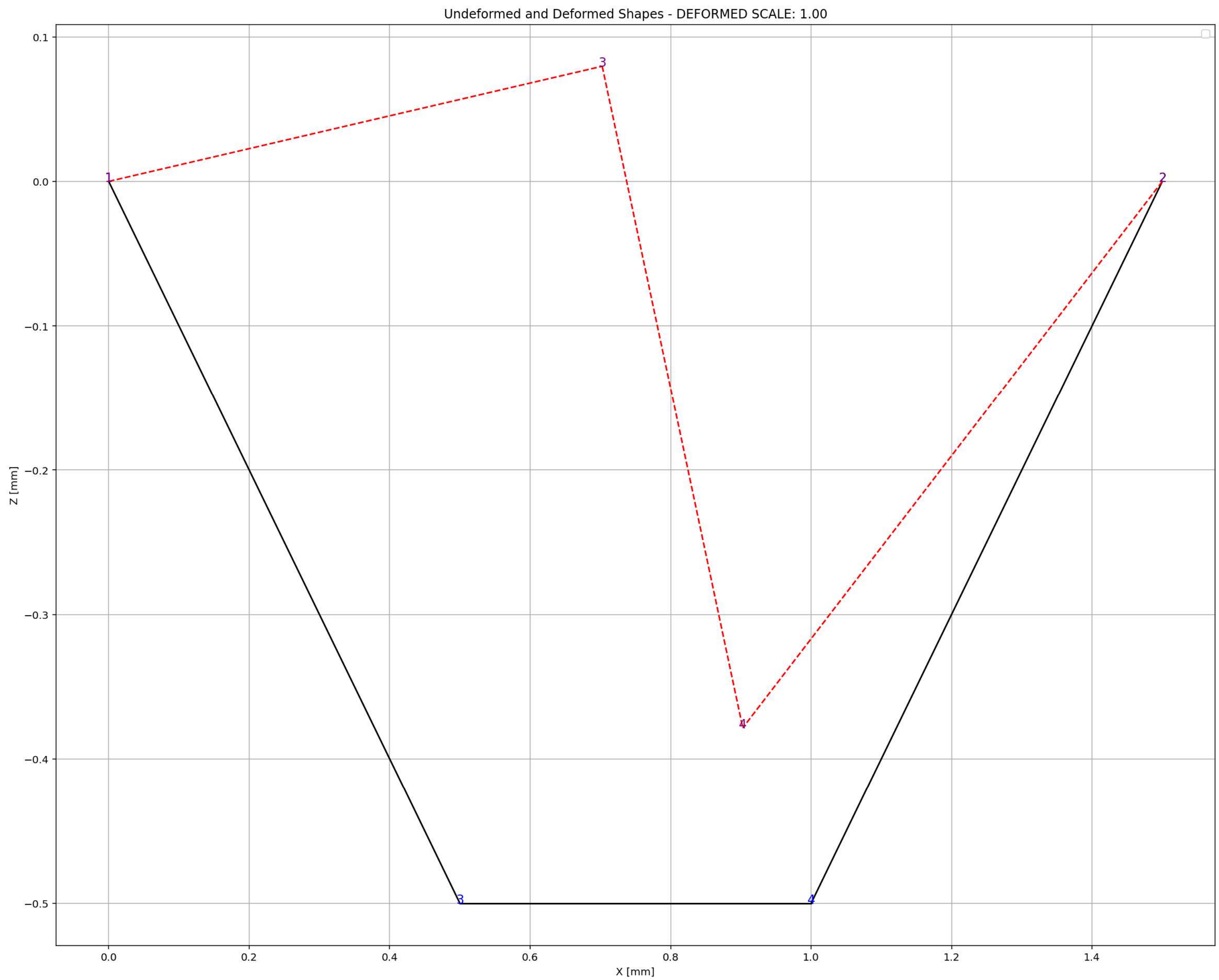
$$P(t) = -m\ddot{u}_g(t)$$

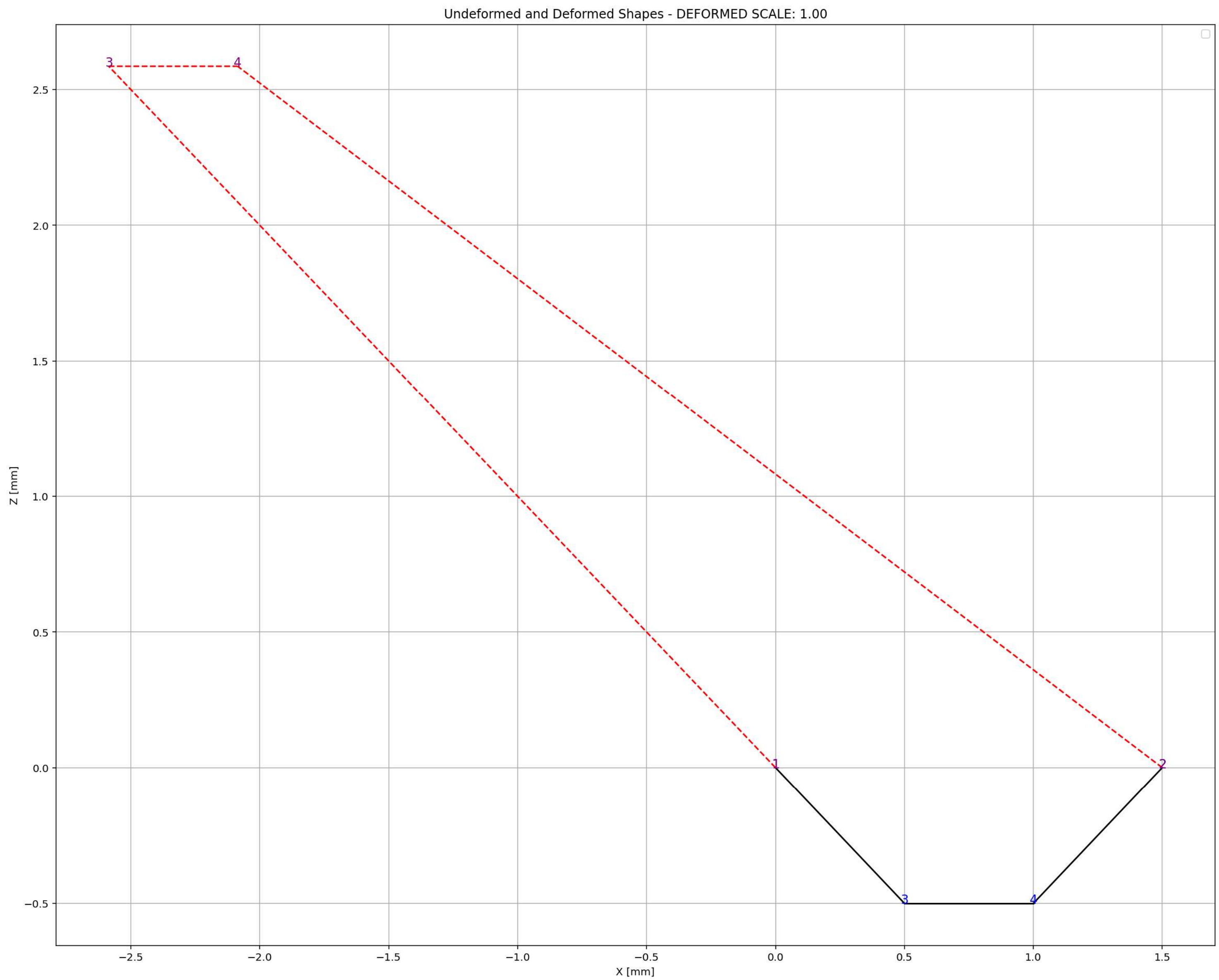

External Time-dependent Loading Over Time [P(t) = -m * ug**(t)]
External Loading - Max: 7.911
Force (N)
Time (s)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
0
1
2
3
4
5

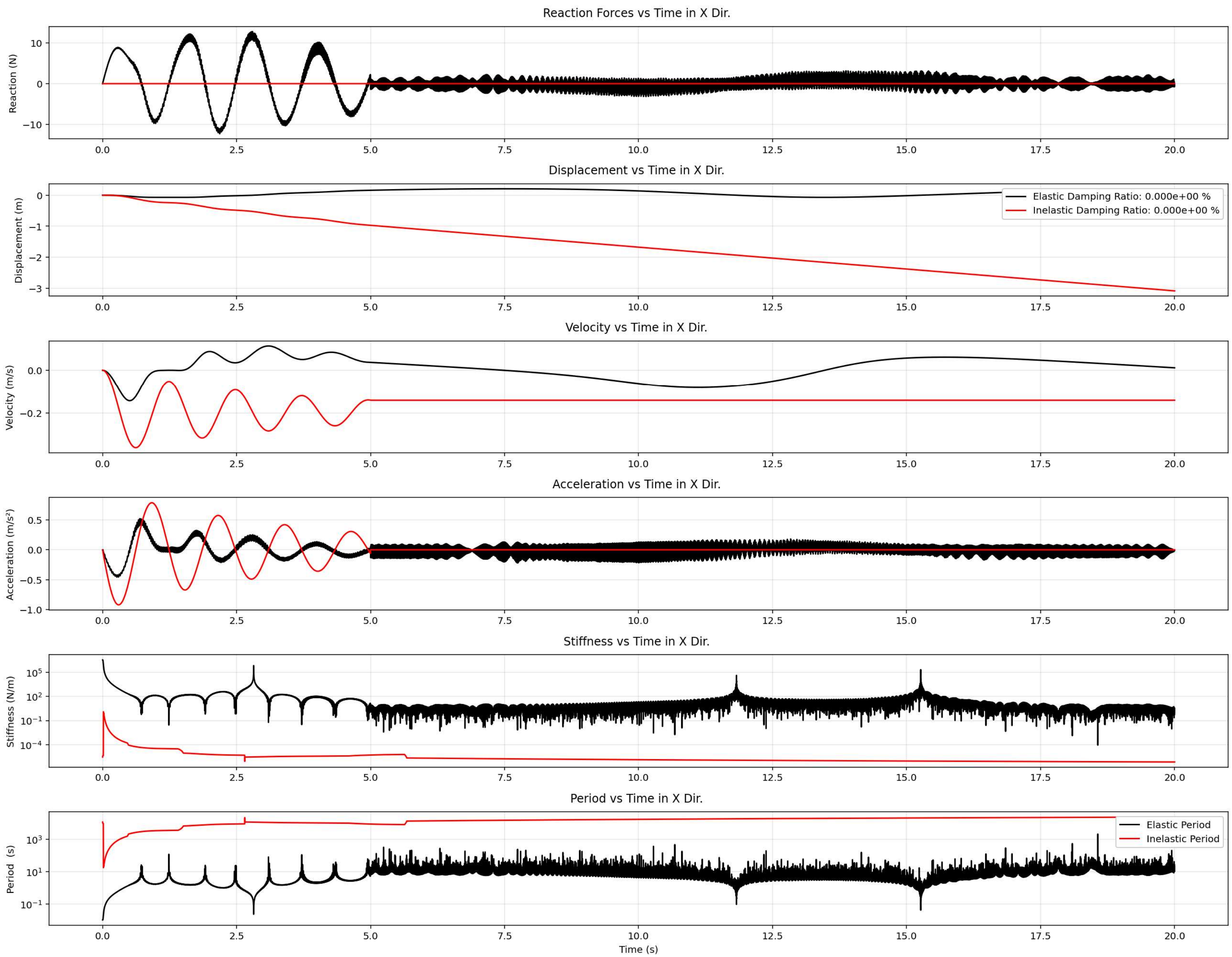


(a) System; (b) stiffness component; (c) damping component; (d) mass component.

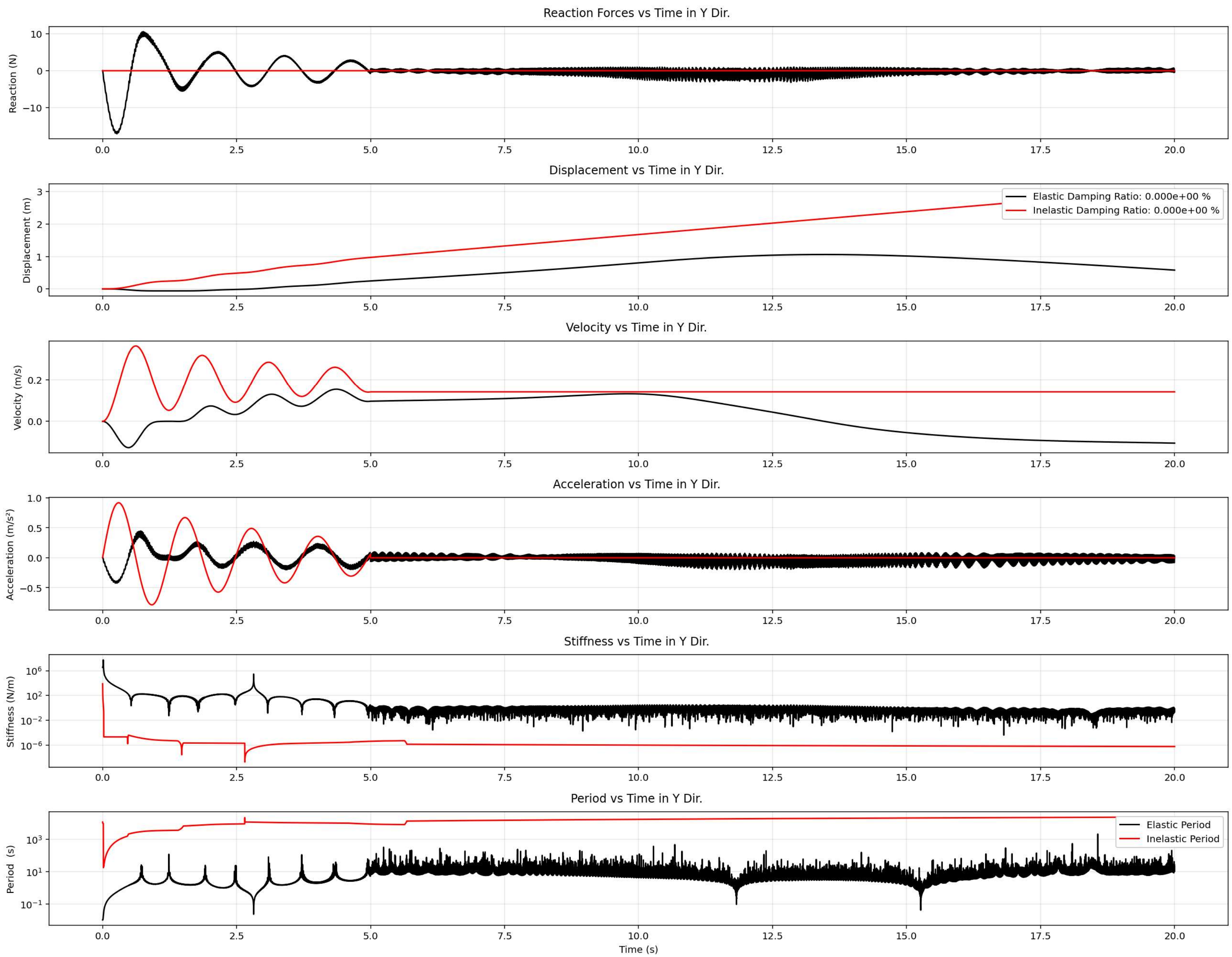



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1
2
3
4
3
4

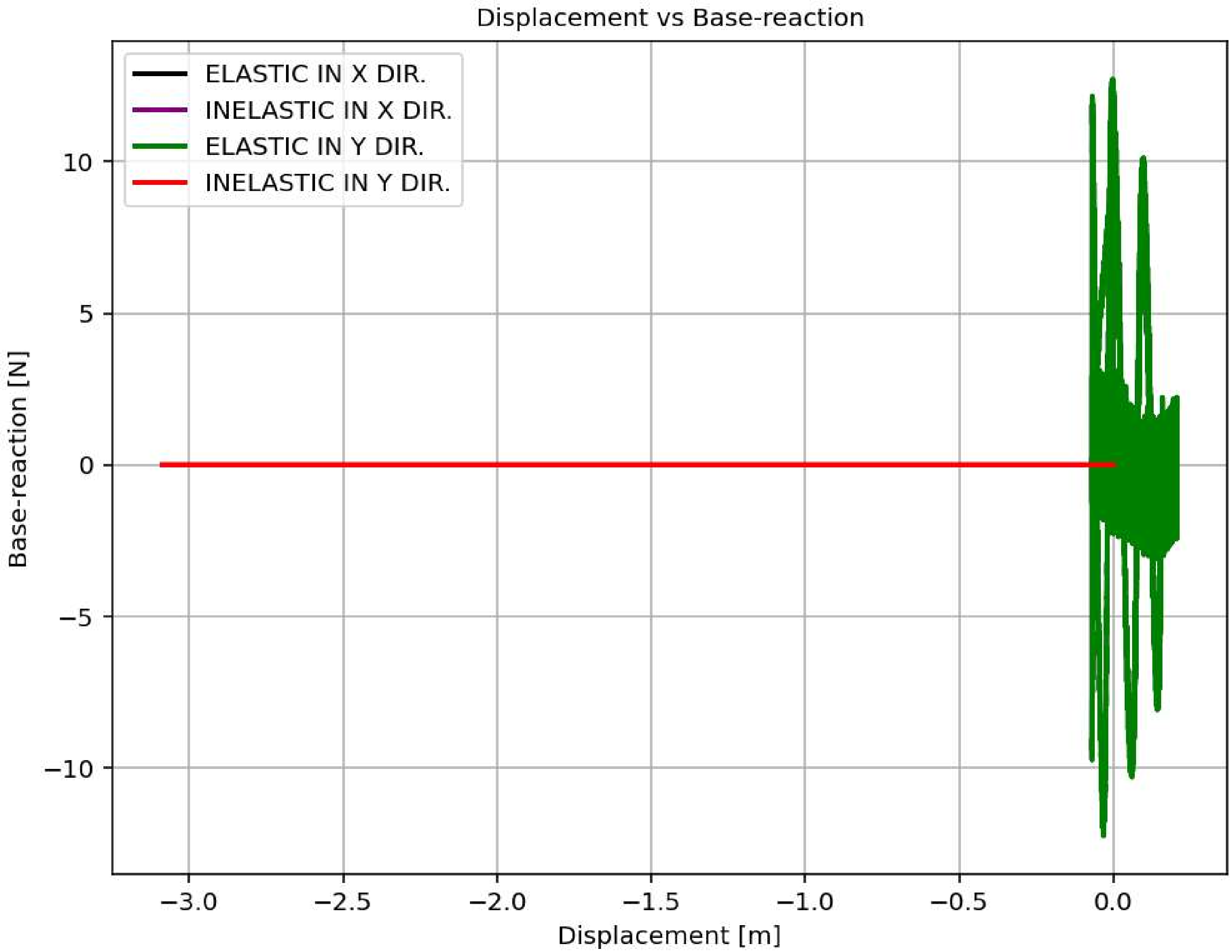

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
1
2
3
4
3
4

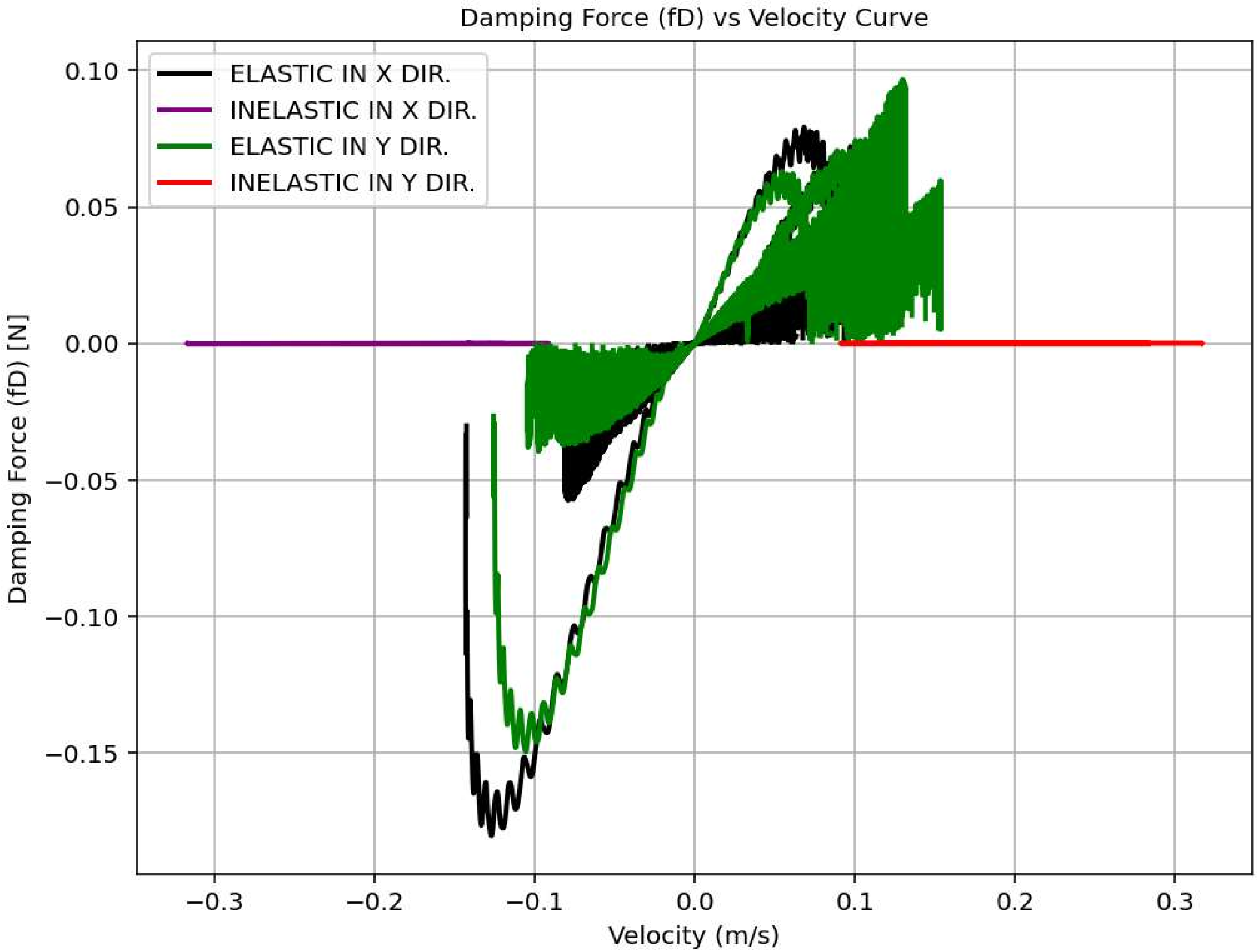


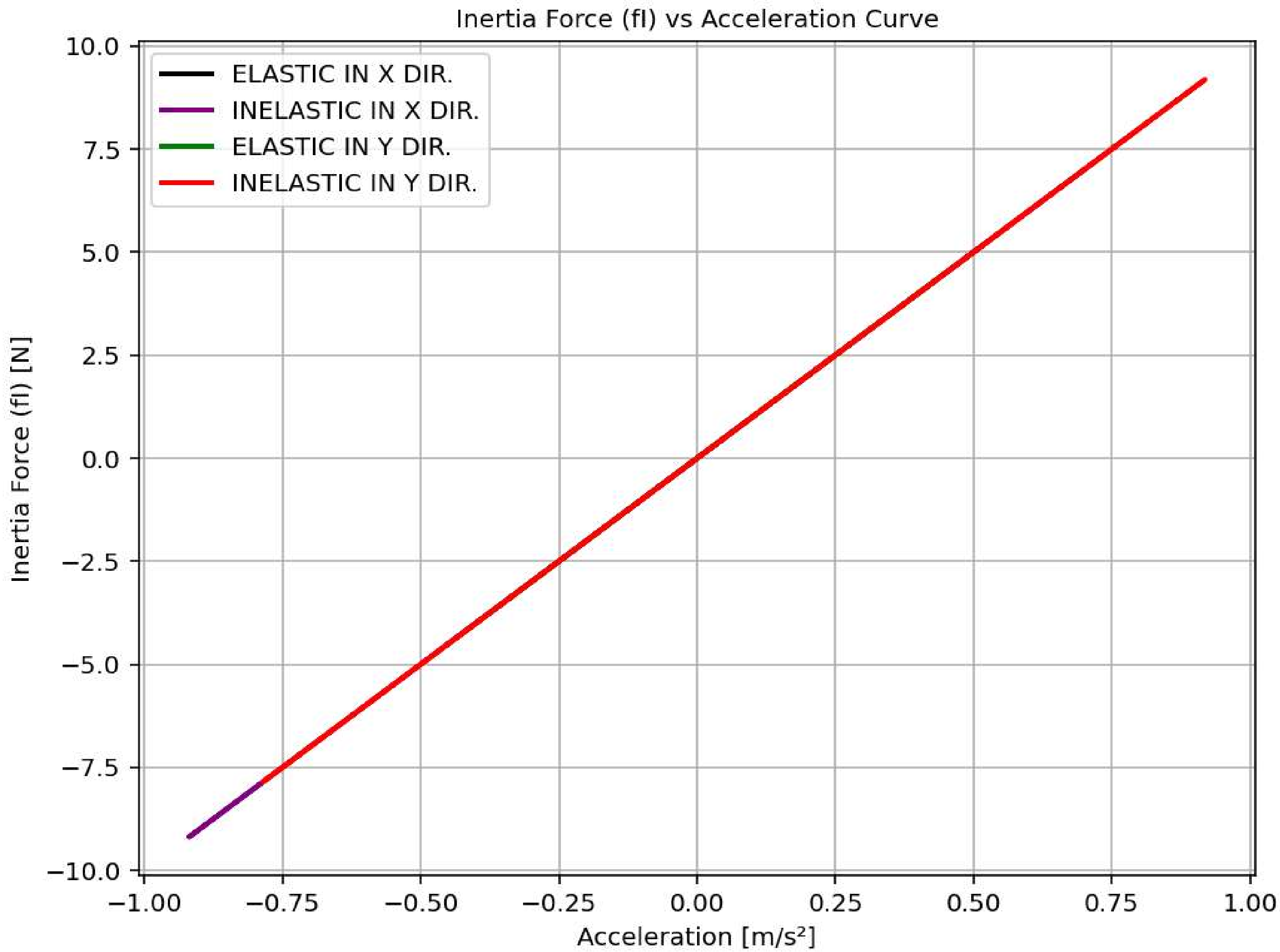
Reaction Forces vs Time in X Dir.
Reaction (N)
Displacement vs Time in X Dir.
Displacement (m)
Elastic Damping Ratio: 0.000e+00 %
Inelastic Damping Ratio: 0.000e+00 %
Velocity vs Time in X Dir.
Velocity (m/s)
Acceleration vs Time in X Dir.
Acceleration (m/s²)
Stiffness vs Time in X Dir.
Stiffness (N/m)
Period vs Time in X Dir.
Period (s)
Elastic Period
Inelastic Period
Time (s)



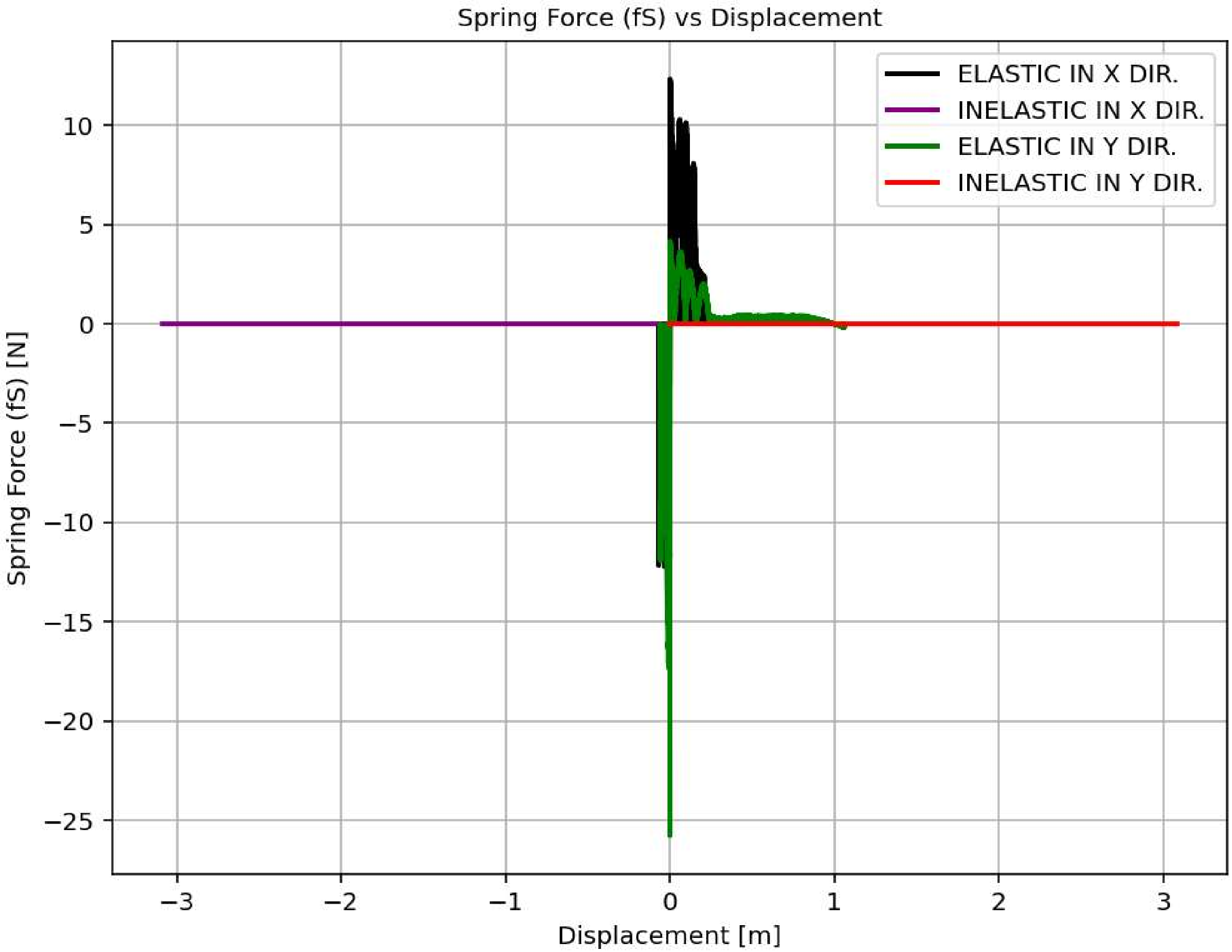
Reaction Forces vs Time in Y Dir.
Reaction (N)
Displacement vs Time in Y Dir.
Displacement (m)
Elastic Damping Ratio: 0.000e+00 %
Inelastic Damping Ratio: 0.000e+00 %
Velocity vs Time in Y Dir.
Velocity (m/s)
Acceleration vs Time in Y Dir.
Acceleration (m/s²)
Stiffness vs Time in Y Dir.
Stiffness (N/m)
Period vs Time in Y Dir.
Period (s)
Elastic Period
Inelastic Period
Time (s)


Displacement vs Base-reaction
ELASTIC IN X DIR.
INELASTIC IN X DIR.
ELASTIC IN Y DIR.
INELASTIC IN Y DIR.
Base-reaction [N]
10
5
0
−5
−10
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Displacement [m]


Damping Force (fD) vs Velocity Curve
ELASTIC IN X DIR.
INELASTIC IN X DIR.
ELASTIC IN Y DIR.
INELASTIC IN Y DIR.
0.10
0.05
0.00
−0.05
−0.10
−0.15
Damping Force (fD) [N]
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Velocity (m/s)



Inertia Force (fI) vs Acceleration Curve
ELASTIC IN X DIR.
INELASTIC IN X DIR.
ELASTIC IN Y DIR.
INELASTIC IN Y DIR.
Inertia Force (fI) [N]
Acceleration [m/s²]
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00



Spring Force (fS) vs Displacement
ELASTIC IN X DIR.
INELASTIC IN X DIR.
ELASTIC IN Y DIR.
INELASTIC IN Y DIR.
10
5
0
−5
−10
−15
−20
−25
Spring Force (fS) [N]
−3
−2
−1
0
1
2
3
Displacement [m]


Damping Force (fD) vs Displacement Curve
ELASTIC IN X DIR.
INELASTIC IN X DIR.
ELASTIC IN Y DIR.
INELASTIC IN Y DIR.
0.10
0.05
0.00
−0.05
−0.10
−0.15
Damping Force (fD) [N]
−3
−2
−1
0
1
2
3
Displacement [m]